

Positive Definiteness of K_d

Setup. Let $\mathbf{d} \in \mathbb{R}^n$ be the Lagrangian finger-space coordinates, $\mathbf{d}_{\text{ref}}$ the virtual equilibrium, and $\delta\mathbf{d} = \mathbf{d}_{\text{ref}} - \mathbf{d}$ the current deflection. The predicted tip force along contact normal $\mathbf{n}$ is

$$\mathbf{f} = \mathbf{n}^\top \mathbf{A} K_d \delta\mathbf{d}, \quad \mathbf{A} = [(\mathbf{P}\mathbf{J}_x)^*]^\top \eta \mathbf{J}_d^\top.$$

Define $\mathbf{v} = \mathbf{A}^\top \mathbf{n}$ (force-sensitivity direction) and $\mathbf{u} = \delta\mathbf{d}$ (deflection), so $\mathbf{f} = \mathbf{v}^\top K_d \mathbf{u}$ and $\gamma = \|\mathbf{v}\|^2 \|\mathbf{u}\|^2$.

Definition 1. K_d is *positive definite* if $\mathbf{x}^\top K_d \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$, equivalently if its symmetric part $K_s := \frac{1}{2}(K_d + K_d^\top)$ is positive definite in the usual sense.

1. Force-tracking update

The gradient of $\mathcal{L} = \frac{1}{2}(f - f_{\text{des}})^2$ is $\partial\mathcal{L}/\partial K_d = e \mathbf{v} \mathbf{u}^\top$, $e = f - f_{\text{des}}$. The update is $K_d^{(t+1)} = K_d^{(t)} - \alpha e_t \mathbf{v} \mathbf{u}^\top$ ($\mathbf{v}, \mathbf{u}$ fixed by the quasi-static assumption).

Convergence and the role of α . Substituting into $f^{(t+1)} = \mathbf{v}^\top K_d^{(t+1)} \mathbf{u}$:

$$f^{(t+1)} = f^{(t)} - \alpha \gamma e_t, \quad \text{so} \quad e_{t+1} = (1 - \alpha \gamma) e_t.$$

The error decays geometrically when $|1 - \alpha \gamma| < 1$, i.e.

$$0 < \alpha < \frac{2}{\gamma}. \tag{1}$$

Summing the *signed* series (geometric, same sign throughout for $\alpha < 1/\gamma$, alternating for $\alpha \in (1/\gamma, 2/\gamma)$, but the sum is always):

$$\alpha \sum_{t=0}^{\infty} e_t = \frac{\alpha e_0}{1 - (1 - \alpha \gamma)} = \frac{e_0}{\gamma}, \tag{2}$$

independent of α . Hence the *converged* stiffness is

$$K_d^{(\infty)} = K_d^{(0)} - \frac{e_0}{\gamma} \mathbf{v} \mathbf{u}^\top,$$

the same matrix for every α satisfying (1).

Theorem 1. Let $K_d^{(0)} \succ 0$, $f_{\text{des}} > 0$, and α satisfy (1). Then $K_d^{(\infty)} \succ 0$ whenever

$$e_0 = f^{(0)} - f_{\text{des}} < \lambda_{\min}(K_s^{(0)}) \cdot \|\mathbf{v}\| \|\mathbf{u}\|. \tag{3}$$

Proof. Let $\mathbf{P} = \frac{1}{2}(\mathbf{v} \mathbf{u}^\top + \mathbf{u} \mathbf{v}^\top)$ (symmetric part of the rank-1 update direction $\mathbf{v} \mathbf{u}^\top$). From (2), $K_s^{(\infty)} = K_s^{(0)} - \frac{e_0}{\gamma} \mathbf{P}$. The largest eigenvalue of $\mathbf{P}$ is $\lambda_+(\mathbf{P}) = (\mathbf{v}^\top \mathbf{u} + \|\mathbf{v}\| \|\mathbf{u}\|)/2 \leq \|\mathbf{v}\| \|\mathbf{u}\|$ (Cauchy-Schwarz). Weyl's inequality applied once to $K_s^{(\infty)} = K_s^{(0)} - \frac{e_0}{\gamma} \mathbf{P}$ gives

$$\lambda_{\min}(K_s^{(\infty)}) \geq \lambda_{\min}(K_s^{(0)}) - \frac{e_0}{\gamma} \lambda_+(\mathbf{P}) \geq \lambda_{\min}(K_s^{(0)}) - \frac{e_0}{\|\mathbf{v}\| \|\mathbf{u}\|},$$

which is strictly positive under (3). For $e_0 \leq 0$ the update adds a positive-semidefinite term to $K_s^{(0)}$, so $K_s^{(\infty)} \succeq K_s^{(0)} \succ 0$ trivially. $\square$

Remark on intermediate iterates. For $\alpha < 1/\gamma$, the error e_t retains its sign throughout (monotone convergence, no overshoot): K_d moves in a single direction from $K_d^{(0)}$ to $K_d^{(\infty)}$. In this case all intermediate iterates also satisfy $K_d^{(t)} \succ 0$, since $K_s^{(t)}$ lies between $K_s^{(0)}$ and $K_s^{(\infty)}$ and both are PD. For $\alpha \in (1/\gamma, 2/\gamma)$ the iterates overshoot and oscillate around $K_d^{(\infty)}$; individual iterates may temporarily lose positive definiteness even if the limit does not.

Key observation. Since $K_d^{(\infty)}$ is the same for every α satisfying (1), reducing α does not change whether $K_d^{(\infty)}$ is PD. The condition (3) depends only on the initial state and the desired force.

Lemma 1 (Isotropic initialization). *If $K_d^{(0)} = k_0 \mathbf{I}$, $k_0 > 0$, condition (3) holds automatically.*

Proof. By Cauchy–Schwarz, $f^{(0)} = k_0 \mathbf{v}^\top \mathbf{u} \leq k_0 \|\mathbf{v}\| \|\mathbf{u}\|$, so $e_0/(\|\mathbf{v}\| \|\mathbf{u}\|) \leq k_0 - f_{\text{des}}/(\|\mathbf{v}\| \|\mathbf{u}\|) < k_0 = \lambda_{\min}(K_s^{(0)})$. $\square$

2. Feedforward stiffness inversion

The first-order tip stiffness is the congruence $K_x = \mathbf{M}^\top K_d \mathbf{M}$, $\mathbf{M} = \mathbf{J}_d(\mathbf{P}\mathbf{J}_x)^*$. Given $K_{x,\text{des}} \succ 0$ and invertible $\mathbf{M}$, inverting yields $K_d = (\mathbf{M}^\top)^{-1} K_{x,\text{des}} \mathbf{M}^{-1}$.

Theorem 2. $K_d = (\mathbf{M}^\top)^{-1} K_{x,\text{des}} \mathbf{M}^{-1} \succ 0$ *always*.

Proof. For any $\mathbf{x} \neq \mathbf{0}$, set $\mathbf{y} = \mathbf{M}^{-1} \mathbf{x} \neq \mathbf{0}$. Then $\mathbf{x}^\top K_d \mathbf{x} = \mathbf{y}^\top K_{x,\text{des}} \mathbf{y} > 0$. $\square$